

July 10, 2026

Editorial Board

Patterns

Cell Press

Dear Editors,

I am submitting “A Pre-Registered Direction Instability Atlas Across Three Perturbation Modalities” for consideration as an Article in *Patterns*.

The paper presents a direction instability (DI) atlas spanning ~55,000 perturbations across three measurement technologies—LINCS L1000 transcriptomics, Tahoe-100M single-cell transcriptomics, and JUMP Cell Painting morphology (CRISPR knockouts and ORF overexpressions)—and characterizes what DI captures through eleven pre-registered hypothesis tests.

DI quantifies whether a perturbation’s effects are consistent or context-dependent across cell types, separating directional variability from effect magnitude. A companion study (Zenodo DOI: 10.5281/zenodo.21213699) established the metric’s validity; this paper asks what DI measures across modalities and perturbation types.

Three results pass their pre-registered criteria: cross-modal concordance confirms DI reflects biology ($\rho = 0.19$, $n = 378$); drug mechanism-of-action class structures DI variation ($\eta^2 = 0.20$, 80 MoA classes); and the essentiality sign-flip—functional constraint suppresses context-dependence—directionally replicates on cross-cell-line CRISPR data ($\rho = -0.26$). Eight failures constrain the metric: expression variance, paralog buffering, and drug selectivity each fail to explain DI variation. A convergent test on DepMap dependency-score variability fails with the wrong sign, exposing a construct divergence between scalar SD and vector cosine DI.

Patterns is a strong fit for three reasons:

1. **Data science methodology applied to biology.** The paper introduces a geometric metric with a formal pre-registration framework that distinguishes passes, failures, and underpowered results. The CI-floor gate that blocks trivially small effects at large n is a general-purpose methodological contribution.
2. **Atlas-scale resource.** The deposit provides magnitude-corrected DI with bootstrap CIs for ~55,000 perturbations, ready for integration with drug-mechanism prediction, gene-function annotation, and screening quality control.
3. **Transparent reporting of negative results.** Eight of eleven tests fail. The failures are as informative as the successes: they delimit the metric and prevent overclaiming. This aligns with *Patterns*’ emphasis on rigorous, reproducible data science.

All atlas data, experiment results, pre-registration documents, and analysis code are deposited on Zenodo (DOI: 10.5281/zenodo.21223667) under CC-BY-4.0. Source code is available at <https://github.com/elliotttower/direction-instability-atlas> under MIT.

I used AI assistants (Claude Code by Anthropic; Perplexity AI) for text editing, code drafting/debugging, and as an independent reviewer of a near-final draft. I reviewed and verified all

outputs and take full responsibility for the content. Full details are disclosed in the manuscript.

I confirm that this manuscript has not been published previously and is not under consideration at another journal. I have no competing interests and this work received no external funding.

Sincerely,

Elliot Tower
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